

Enterprise EIGRP WAN Network Implementation Project

Detailed Project Implementation & Summary Report

This report documents the complete design, implementation, configuration, testing, troubleshooting, and verification process of the Enterprise EIGRP WAN Network Project. The project was designed as a practical enterprise-level networking implementation focused on redundancy, failover, load balancing, and dynamic routing using Cisco technologies.

1. Project Topology Overview

The topology consisted of one Headquarters (HQ) region and three branch offices connected through multiple ISP routers. Each branch contained two routers connected redundantly to multiple ISP routers in order to simulate a real enterprise WAN environment. The WAN routing protocol used was EIGRP. Redundancy and failover mechanisms were implemented using Equal-Cost Load Balancing (ECMP), EIGRP failover, and VRRP-based gateway redundancy.

Location	LAN Network	Routers	Connected ISPs
HQ - Bangalore	10.0.1.0/24	HQ_ROUTER_1, HQ_ROUTER_2	ISP1 - ISP6
Branch1 - Kolkata	192.168.1.0/24	BRANCH_1_R1, BRANCH_1_R2	ISP1, ISP2
Branch2 - Assam	192.168.2.0/24	BRANCH_2_R1, BRANCH_2_R2	ISP3, ISP4
Branch3 - Delhi	192.168.3.0/24	BRANCH_3_R1, BRANCH_3_R2	ISP5, ISP6

2. Step-by-Step Project Implementation

STEP 1 — Topology Design

The first step was designing the enterprise WAN topology. The topology was designed with one HQ region and three branch locations. Multiple ISP routers were included in order to simulate real enterprise WAN redundancy and multi-homing.

Problem Solved:

- Created a scalable enterprise WAN design
- Introduced redundancy and high availability
- Allowed future failover and load balancing implementation

STEP 2 — LAN IP Addressing

LAN IP addressing was configured for all branch locations and the HQ region. Each branch was assigned a

separate subnet while maintaining structured IP addressing for routers and end devices.

Problem Solved:

- Established communication within each branch LAN
- Created proper subnet segmentation
- Enabled structured enterprise addressing

STEP 3 — WAN IP Addressing

WAN IP addressing was configured between branch routers, ISP routers, and HQ routers using multiple WAN subnets.

Problem Solved:

- Established router-to-router communication
- Enabled ISP transit routing
- Allowed EIGRP neighbor formation later in the project

STEP 4 — Connectivity Verification

After IP addressing, direct connectivity testing was performed using ping commands between routers, ISPs, and end devices.

Problem Solved:

- Verified Layer 3 connectivity
- Identified interface and IP addressing issues early
- Confirmed WAN links were operational

STEP 5 — EIGRP Configuration

EIGRP AS 100 was configured on all HQ routers, branch routers, and ISP routers. Network statements were added to advertise all LAN and WAN networks.

Problem Solved:

- Enabled dynamic route exchange
- Removed the need for static routing
- Allowed all branches to communicate dynamically

STEP 6 — EIGRP Neighbor Verification

The command **show ip eigrp neighbors** was used to verify EIGRP neighbor adjacency formation between all routers and ISPs.

Problem Solved:

- Confirmed proper EIGRP operation
- Verified successful neighbor formation
- Ensured route exchange was occurring correctly

STEP 7 — Branch-to-Branch Communication Testing

Ping and traceroute testing were performed between branches and HQ devices.

Problem Solved:

- Verified end-to-end enterprise connectivity
- Confirmed successful dynamic routing
- Validated ISP transit routing paths

STEP 8 — Equal-Cost Load Balancing (ECMP)

It was observed that EIGRP automatically installed multiple equal-cost routes because WAN metrics between ISP links were equal.

Problem Solved:

- Enabled traffic load balancing across WAN links
- Improved WAN bandwidth utilization
- Introduced redundancy automatically

STEP 9 — Automatic Failover Testing

WAN interfaces connected to ISPs were shut down manually to simulate ISP failure scenarios.

Problem Solved:

- Verified automatic failover functionality
- Demonstrated EIGRP reconvergence
- Confirmed high availability of WAN connectivity

STEP 10 — Unequal-Cost Load Balancing using Variance

Delay manipulation and EIGRP variance were tested on Branch2 in order to demonstrate unequal-cost load balancing concepts.

Problem Solved:

- Demonstrated feasible successor concepts
- Showed metric-based path selection

- Implemented unequal-cost load balancing behavior

STEP 11 — VRRP Configuration

VRRP was configured on all LAN-facing interfaces between R1 and R2 routers in each branch and HQ location. Virtual IP addresses were assigned for gateway redundancy.

Problem Solved:

- Eliminated single point of gateway failure
- Implemented first-hop redundancy
- Allowed seamless gateway failover between routers

STEP 12 — VRRP Failover Verification

Primary router interfaces were shut down to verify VRRP failover operation.

Problem Solved:

- Verified gateway redundancy functionality
- Confirmed automatic master/backup transition
- Demonstrated enterprise LAN resiliency

3. Features Successfully Implemented

- Enterprise WAN topology design
- Multi-ISP WAN architecture
- Dynamic routing using EIGRP
- Branch-to-branch communication
- HQ-to-branch communication
- Equal-Cost Load Balancing (ECMP)
- EIGRP automatic failover
- EIGRP feasible successor concepts
- Unequal-Cost Load Balancing using variance
- WAN redundancy and high availability
- VRRP-based first-hop redundancy
- Gateway failover implementation
- Enterprise troubleshooting and verification

4. Troubleshooting & Verification Commands

The following troubleshooting and verification commands were used during the implementation process:

- show ip interface brief
- show ip route
- show ip eigrp neighbors
- show ip eigrp topology
- show vrrp
- ping
- traceroute

These commands helped verify:

- Interface status
- Route learning
- EIGRP neighbor relationships
- Load balancing behavior
- Failover operation
- VRRP master/backup status

5. Project Conclusion

The Enterprise EIGRP WAN Network Project was successfully designed and implemented. The topology demonstrated enterprise-level routing concepts including redundancy, failover, load balancing, dynamic routing, and gateway resiliency. The project provided strong hands-on exposure to real-world enterprise networking concepts and significantly improved understanding of EIGRP routing behavior, WAN redundancy, VRRP operation, and enterprise troubleshooting methodologies. As a first enterprise-level networking implementation project, the topology successfully achieved all major design objectives and demonstrated a strong practical foundation in enterprise routing and WAN technologies.